When Does Verification Help? Recovery-Path Regularization for Strategy-Aware Valuation

Yuchen Tang Fengqin You* Junxiang Gu Xin Pang

School of Computer and Artificial Intelligence
Nanjing University of Science and Technology Zijin College
Nanjing, China

*Corresponding author: irisyou@njust.edu.cn

Abstract

Recovering withheld information can increase a seller's predicted price when the valuation model is nonmonotone. We study this failure under natural missingness, charged failed queries and shared verification capacity. A recovery-path decomposition separates information recovery from its effect on concealment gains. We introduce recovery-path regularization, which penalizes increases in positive concealment gain along legal recovery edges while retaining public-record deployment inputs. A restricted information frontier states the limits of verification, rather than predicting learned-model performance. In a matched ten-seed study at 10% query capacity, the declared unit-weight endpoint reduces mean inflation by 50.41%, 26.58% and 23.81% on synthetic, JUCars and MUCars relative to same-gap zero-weight controls. All thirty source–seed comparisons have the same direction. Synthetic honest mean absolute error rises 0.82% and failed queries rise 1.10 percentage points, so lower inflation does not imply multiobjective dominance. Stronger training-response search also removes some earlier feedback damage, showing that the negative feedback result depends on implementation. On the real sources, most improvement is already present without queries, limiting a recovery-mechanism interpretation. A separate three-seed, 90-cell crossing lowers inflation in all 45 paired comparisons when RPL is combined with five task-adapted acquisition families. Four-stage checks cover 1,206 earlier extension cells; the crossing has independent outcome checks. Remaining response regret, retrospective cohorts and cross-machine numerical differences bound the findings. These retrospective fixed-cohort results support a proof of concept for evaluating concealment gains jointly with prediction and verification costs.

Keywords: strategic missingness; secondhand valuation; budgeted verification; incentive auditing; factorial evaluation

1 Introduction

Secondhand valuation combines incomplete seller reports with scarce verification. Natural absence and deliberate withholding can produce the same public record; an attempted query may fail while still consuming capacity. Even successful recovery need not improve incentives. A nonmonotone predictor can quote 10 for an honest record, 12 after two fields are hidden, and 14 after one is recovered. These illustrative quote units expose the distinction between recovering information and reducing the seller's gain. Shared batch capacity adds another dependence: changing one report can displace another listing from the query set.

Strategic-missingness regression and classification [10, 6], budgeted audits [4] and partial verification [7] provide established approaches under their respective outcome models. Predictive feature acquisition instead targets information or error reduction [13]. None directly guarantees that a finite feedback-training implementation will improve continuous quotations after failed-but-charged recovery. Our outcomes are advertised-price predictions and simulated seller responses, not observed transactions or a transfer mechanism.

We link recovery-path diagnosis to a targeted training intervention. First, an information frontier isolates the limits imposed by uniform honest precision and observationally identical missing reports. Its restricted family does not predict a learned model's empirical optimum. Second, a signed recovery decomposition exposes gain-increasing intermediate records in fitted four-field models. Recovery-path regularization (RPL) penalizes those increases without adding private deployment inputs. Third, matched experiments separate initial objective, acquisition, feedback and allocation. We assess RPL with ten fitting seeds and a separate 90-cell crossing with five recent acquisition families, including native-head references and prediction/query costs. Component removals, budget variation and computational checks bound both studies.

The evidence supports conditional use. At the declared unit-weight endpoint, RPL lowers inflation in all thirty source–seed comparisons, but synthetic honest error and query failures rise. Stronger training-response search removes some earlier feedback damage, restricting that negative finding to its implementation. Competitive simple controls, finite-response regret and cross-machine numerical differences further bound the claims. The practical question is therefore which recovery-aware intervention improves the joint incentive, prediction and verification outcome for a stated setting.

2 Related work

Withholding and verification. Support-margin imputation addresses strategic missingness in regression [10]; truthful classifiers address withholding in classification [6]. Our precision frontier adds a uniform honest-quote constraint to a hard query budget, rather than introducing ambiguous missingness as a new problem. Audit and partial-verification mechanisms [4, 7] motivate checking reports, while established marginal representations [3] permit compact finite allocation. Their guarantees remain specific to their mechanism classes.

Monotonicity regularization and certification predate RPL [5]. Where that approach regularizes input-coordinate derivatives and certifies monotonicity, RPL applies a soft constraint to finite recovery edges of honesty-relative positive gain. It offers no global monotonicity certificate. The contribution is this recovery-graph instantiation and its empirical trade-offs, not a new regularization principle or a truthfulness proof.

Acquisition and strategic feedback. ACO [13], amortized greedy selection (GDFS) [18], discriminative information estimation (DIME) [19], stochastic latent encodings (SEFA) [20] and explanation-ranked acquisition [21] broaden the acquisition comparison beyond risk and predictive benefit. We run task adapters, with boundaries in Appendix C; they are not official end-to-end reproductions. Performative prediction and strategic risk minimization [11, 12] study deployment-dependent responses; contextual games and audit-risk design [8, 9] condition decisions on available information. We neither transfer their convergence guarantees to our finite nonconvex loop nor claim randomization or monotonicity as new principles. The contribution is the recovery-targeted loss and its controlled evidence, including search-strength interventions that qualify feedback failure.

3 Problem formulation

3.1 Reports, queries and allocation

A batch contains $n = 100$ listings. Listing i has recorded vector x_i and natural missingness M_i. A seller conceals at most two available fields among mileage, transmission, fuel type and body condition. Its report $r_i(s_i)$ marks $M_i \cup s_i$ missing; values cannot be fabricated or recovered through identifiers. Missingness indicators do not distinguish natural absence from concealment. Reports are locked before private allocation randomization.

The platform may query at most one publicly missing field per listing and at most $B = \lfloor bn \rfloor$ fields on every draw. A concealed recorded value is recovered; natural absence remains unavailable. Both attempts spend capacity. Deployment scores use public records and model outputs, not latent recoverability or advertised targets. Seller utility is expected log quote minus concealment cost, zero in the main analysis. Response search reallocates the whole batch for each candidate own report with peers fixed.

SAV combines an augmented predictor, suppression scoring and query-feedback training. Static robust changes the prediction objective without that feedback. CDM fixes the price model and calibrates a mixture of twelve feasible rules on stored report/peer rows from five profile families; it has no unseen-type guarantee. The matched factorial below trains new initial-objective fits, rather than renaming these original checkpoints.

Let q_s be a cross-fitted report quote, $q_{s\setminus f}$ its quote after recovering f, q^H the honest quote and y the log target. Acquisition labels are

$$v_{sf}^{\text{sup}} = [q_s - q^H]_+ - [q_{s\setminus f} - q^H]_+, \quad v_{sf}^{\text{ben}} = (q_s - y)^2 - (q_{s\setminus f} - y)^2, \quad v_{sf}^{\text{risk}} = (q_s - y)^2.$$

Common ExtraTrees heads [15] learn labels from public encodings; honest records and targets are training supervision only. Each policy picks its highest positive eligible field. Top-B ranks item scores with stable ties. Score-minimax minimizes $\max_i g_i(1 - p_i)$ over query marginals $0 \le p_i \le 1, \sum_i p_i \le B$, where g_i is the selected positive score. This is an additive score surrogate, not the true report-dependent incentive objective; every changed report therefore triggers reallocation.

3.2 Joint outcomes and finite responses

Write $y = \log(1+\text{price})$, let q_0, q_1 be pre-query and selected-query quotes and p the query marginal. With $\mu = (1-p)q_0 + pq_1$, inflation is $[\mu - q^H]_+$. A relative reduction of this metric is neither the same percentage price reduction nor observed overpayment avoided. It differs from the realized-positive quantity $(1-p)[q_0 - q^H]_+ + p[q_1 - q^H]_+$. Honest and attacked MAE are respectively

$$|e^{q^H} - e^y|, \qquad (1-p)|e^{q_0} - e^y| + p|e^{q_1} - e^y|.$$

Averaging quotes before absolute error would define another metric. Failed-query probability is p times the indicator of unavailable recovery, aggregated *per all listings*; failure still spends capacity. Signed log bias is $\mu - y$. Inflation, attacked MAE and failed queries are co-primary descriptive outcomes, with honest MAE measuring precision. This retrospective choice guards against a constant predictor, which has zero inflation regardless of pricing error; it is not a preregistered endpoint change.

The unilateral deviation envelope is $[\max_r u(r) - u(r_{\text{honest}})]_+$ with fixed peers and fresh batch allocation. Remaining regret subtracts $u(r_{\text{returned}})$ instead. Small regret does not imply small gain over honesty or a globally selected market equilibrium. All returned profiles, including nonconverged ones, remain in aggregation.

4 Information limits and finite comparators

4.1 An attainable precision frontier

Consider a missingness-alias family: an all-natural-missing world and worlds in which only item i has a recoverable low-value field lowering its target by $\Delta_i > 0$. Withholding produces the same public batch. Quotes are itemwise and verification-neutral: identical final records receive identical quotes. Optimizing both the quote function and verification under uniform honest absolute error η, set $a_i = (\Delta_i - 2\eta)_+$. The attainable worst positive expected gain is

$$T^*(\eta, B) = \min_{0 \le p_i \le 1,\ \sum_i p_i \le B} \max_i a_i(1 - p_i). \tag{1}$$

For a nonzero interior optimum, $\sum_{i:a_i>0}(1 - T^*/a_i)_+ = B$.

Proof and scope. Couple policies on the private random seed along the all-failure transcript. The probability p_i of querying item i has $\sum_i p_i \le B$. Until that query, its concealed-low world has the same transcript. Uniform precision on both honest worlds leaves a gap at least $\Delta_i - 2\eta$; verification-neutrality prevents unsuccessful queries changing the quote. This proves the lower bound, including adaptive policies in this family. Shift the two quotes toward one another by $\min(\eta, \Delta_i/2)$ and sample the optimizing marginals under hard capacity to attain it. Thus $T^* = 0$ exactly when $B \ge |\{i : \Delta_i > 2\eta\}|$; for identical gaps, $T^* = (\Delta - 2\eta)_+(1 - \min\{B/n, 1\})$.

The 2η term comes from satisfying two honest reports with one quotation rule. It does not follow from average MAE or contradict truthful classifiers that change predictions without this precision constraint [6]. Our learned four-field models violate the frontier's single-recoverable-field and uniform-error assumptions. We therefore use it to delimit guarantees, not to predict empirical feedback signs or model optima.

4.2 Compact finite oracle

For each public batch class o, the finite oracle shares one action lottery across all compatible latent worlds and minimizes the largest expected unilateral gain, recomputing truthful and deviated allocations. For item-local utility, unit-cost nonadaptive queries and at most one field per item, it can instead use field marginals y_{oif} satisfying

$$y_{oif} \ge 0, \qquad \sum_f y_{oif} \le 1, \qquad \sum_{i,f} y_{oif} \le B,$$

with zero mass on ineligible fields. Every deviation is affine in these shared marginals, including both truthful and deviated public classes.

Proposition 1 (Compact implementability)**.** *The marginal and complete-allocation finite oracles have the same value in this class, and every feasible marginal solution respects capacity on each draw.*

Proof. A complete lottery induces feasible marginals. Conversely set $p_i = \sum_f y_{if}$ and concatenate half-open intervals of lengths p_i. For uniform $U \in [0, 1)$ select intervals containing some $U + k$, integer k. Each length is at most one and their total at most integer B, so at most B items are selected, with probabilities p_i. Given selection, choose field f with probability y_{if}/p_i. This realizes all marginals and preserves every affine expected payoff. □

Marginal optimization is established [3]. This specialization uses $OnF + 1$ variables but does not remove world/report enumeration, preserve arbitrary joint-event probabilities or cover general adaptive/nonunit-cost queries. Certificates cover the declared finite universe only.

Compact and complete values agree on 32 original and 24 multi-field fixtures; all 57 primal/dual certificates pass. A larger $n = 100, F = 4, O = 8, B = 10$ fixture uses 3,201 variables versus 18,662,296,388,649,543,921 complete allocations per class. Full enumeration was not run there, so this is a representation count, not measured speedup. At $B = 1, \eta = 0$, the two-mode toy library (six three-item priorities, risk and no query) differs from empirical CDM's twelve rules. Its 72/17 oracle, 72/13 global-mixture and 8 risk values use artificial quote units. Appendix A gives a contextual separation and the stronger $B + 1$-rule comparator that closes it.

5 Recovery diagnosis and recovery-path learning

5.1 Separating recovery from competition

For fixed peers and an evaluated mixture, retain its maximizing own report s^*. Let q_h be the unqueried report quote, q_L the quote after the same rule's highest positive eligible query with capacity relaxed, and q_A the hard-budget quote. At the same report,

$$E = q_h - q^H, \quad R = \mathbb{E}[q_L] - q_h, \quad C = \mathbb{E}[q_A] - \mathbb{E}[q_L], \quad G = E + R + C.$$

All terms are signed; mixture expectation precedes report maximization. The relaxed query is diagnostic. Reusing the truthful query would target a naturally missing field and confound recovery with competition. In the 10/12/14 example, $E = R = 2$, $C = -2(1-p)$ and $G = 2 + 2p$: successful partial recovery can increase the gain of a nonmonotone predictor.

5.2 Recovery-path regularization

Let H_i be record i's honest report and $\mathcal{E}_i$ its legal edges recovering actually concealed fields; natural unavailability is excluded. For quoted log price q_θ, define

$$g_{i,\theta}(s) = [q_\theta(s) - \mathrm{sg}(q_\theta(H_i))]_+, \tag{2}$$

$$L_{\mathrm{path}}(\theta) = \frac{\sum_i \sum_{(s,a)\in\mathcal{E}_i} [g_{i,\theta}(T(s,a)) - g_{i,\theta}(s)]_+}{\sum_i |\mathcal{E}_i|}. \tag{3}$$

RPL adds $\lambda L_{\mathrm{path}}$ to its matched no-feedback loss. The existing gap term penalizes gain over honesty; this term penalizes an *increase* in that positive gain along recovery. The honest anchor is detached within an optimizer step, although shared parameters can move it between steps. Honest values supply training supervision only. Deployment retains the public-record predictor without private recoverability or quotation history.

Proposition 2 (Finite recovery-graph implication)**.** *If $L_{\mathrm{path}} = 0$ on fixed finite record graphs, positive gain cannot increase along any path of their legal recovery edges.*

Proof. Each summand is nonnegative. A zero sum forces every edge increment to be nonpositive; chaining these inequalities proves the claim. □

The implication permits positive initial concealment gain and says nothing about unseen records, finite-penalty fits, honest error or allocation optimality. Because $g_{i,\theta}(H_i) = 0$, an edge ending at honesty has zero penalty regardless of its starting gain. The constraint acts on intermediate partial-recovery states. It motivates a recovery-graph-inspired regularizer, not a general incentive guarantee.

5.3 Conditional cap and final-record alternative

A diagnostic history cap uses $q_{\mathrm{cap}}(r,a) = \min\{q(r,\varnothing), q(r,a)\}$ before mixture averaging, fixing the model, public scores and rule weights. If truthful queries leave q^H unchanged, then pointwise $q_{\mathrm{cap}}(s,a) \le q(s,a)$ while $q_{\mathrm{cap}}(H,a) = q^H$. Subtracting q^H, averaging and maximizing over legal own reports proves that the worst fixed-peer gain weakly decreases. This is not a guarantee after peers recompute responses, nor a guarantee of lower absolute error.

The cap can quote identical final records differently because it retains history, and lies outside the verification-neutral frontier class. An alternative is $q_{\downarrow}(r) = \min_{z \in \mathcal{C}(r)} q(z)$ on a fixed completion domain. If completion sets are nonempty and nested and cover the truthful record, then $\mathcal{C}(r^H) \subseteq \mathcal{C}(r)$ implies $q_{\downarrow}(r) \le q_{\downarrow}(r^H)$, eliminating gains over its own honest quote with nonnegative costs. Unlike the history cap, it can reduce honest quotes and worsen MAE. Unsupported records need abstention or an uncertified fallback. This envelope is related to established truthful withholding prediction [6, 10]; a restricted empirical cost pilot is reported in Appendix B. Equal-record consistency, appeals and possible selection disparities require explicit platform review; we have not measured protected-group effects or deployment fairness.

6 Experimental design and computational checks

6.1 Cohorts and role-separated fitting

JUCars-2024 v2 contains Jordanian listings from 2024, deposited in 2026 [1]; MUCars-2024 v2 contains Moroccan 2024 listings [2]. Both use CC BY 4.0 and advertised, not transaction, prices. Native currencies are evaluated separately. Processed training/validation sizes are 4,774/1,374 and 39,244/10,672. Label-independent identifier hashing caps training at 5,000 and evaluation at 1,000. Synthetic seeds 73,101/73,102 generate 6,000/1,500 records before the same caps. All fitting seeds share these populations and ten fixed batches of 100. Real cohorts were inspected during development; these are retrospective comparisons. Inherited non-price fingerprints control duplicates without certifying unique vehicles. Targets, identifiers, source indices and declared reveal aliases are excluded; correlated features remain. Fields unavailable in the source record are stipulated to be unrecoverable in the simulation; their real missingness causes are not identified. They cannot be additionally concealed. Supplement S1 reports the complete sample flow, field mappings, missingness and synthetic generator.

The original factorial crosses two initial objectives, three acquisition scores, feedback on/off and two allocators: 24 cells per source/seed/capacity. Sharing no-feedback fits gives fourteen unique fits. Every outer fold is excluded from its entire fitting loop; the remaining records are split into estimator and 30% guidance partitions. A 250-round CatBoost backbone [14] is fitted with valid-mask augmentation. Residual branches start from this same backbone without a fitted neural residual. Original feedback branches use two six-epoch updates; the original static no-feedback branch uses one twelve-epoch call. Matched RPL and stronger-gap branches use two six-epoch updates with identical minibatch orders and optimizer resets. Supplement S1 distinguishes the experiment families. Guidance heads refresh after each round. At each feedback epoch's start, public scores are fixed; within minibatches the current detached predictor searches responses with one start and two sweeps at 10% capacity. No-feedback uses no-query response search with the same update budget. Deployment heads use assembled excluded-fold targets and deployment encodings; the final predictor is not updated after that head fit. No model is refitted for another evaluation capacity.

Writing L_H, L_V, L_A for honest squared error, valid-report mean squared error and backbone-

honest anchoring, mask training uses $L_H + .25L_V + .25L_A$. Robust training adds $.5L_{\text{attack}} + .5L_{\text{gain}}$, with $L_{\text{gain}} = \mathbb{E}[(\mu - \text{sg}(q^H) - .02)_+]$. The RPL comparisons vary their declared gap/path weights under the same no-feedback reset schedule. Endpoint-error training uses

$$(1-p)(q_0-y)^2 + p(q_1-y)^2 = (\mu-y)^2 + p(1-p)(q_1-q_0)^2,$$

not squared error after averaging. Gradients do not pass through report search, field choice or allocation. The residual has two width-64 ReLU layers; AdamW uses learning rate .001, weight decay 10^{-4} and gradient clipping at 5. Table 1 collects settings and the separate study scopes.

Setting	Value
CatBoost backbone boosting rounds [14]	250
Residual hidden width / learning rate	64 / 0.001
RPL residual epochs / optimizer-reset rounds	12 / 2
Outer folds / guidance fraction	3 / 0.3
Acquisition-head trees / maximum depth / minimum leaf	160 / 12 / 12
Training budget / response starts / response sweeps	10% / 1 / 2
Evaluation response starts / maximum sweeps	4 / 20
Batch size / maximum concealed available fields	100 / 2
Gain-penalty tolerance / main concealment cost	0.02 / 0
Original factorial seeds / bootstrap repetitions	10 / 1,000
RPL path weight λ / gap weight β	$0, 0.1, 0.5, 1$ / 0.5
RPL gap-removal path weights / gap	$0, 0.5$ / 0
RPL evaluation query capacities	$0, 0.1, 0.3$
Completion / RPL-extension fitting seeds	3 / 10
Core-extension audit resamples	4,000
Recent acquisition families / frozen price backends	5 / 2
Recent-method prediction bins	4
ACO neighbor default / fixed sensitivity values	25 / 10, 25, 50
Additional natural-missingness fractions	$0, 0.1, 0.3$

Table 1: Declared settings across experiments. The ten fitting seeds are 13, 42, 2026, 101, 211, 307, 401, 503, 601 and 701. The backbone and ExtraTrees implementations follow established methods; they are not new algorithmic contributions. All capacities share the fixed training budget. Three-seed completion experiments use the first three seeds; RPL extends to all ten. The additional settings distinguish original-factorial inference from the new audit families.

6.2 Evaluation, uncertainty and audit scope

Fixed models are evaluated with four starts and at most twenty response sweeps; quotes/errors are integrated analytically over query marginals and every legal unilateral deviation is checked with fresh reallocation. The original study has 4,320 cells over thirty source–seed pairs. Its 58 conditional contrasts per source/capacity produce 174 primary contrasts. Paired crossed seed–batch resampling preserves method pairing [16]: the original analysis uses 1,000 resamples, whereas the core extensions use 4,000, including exploratory studentized maximum-deviation bands [17] over 216 RPL metric contrasts. Fixed cohorts, ten clusters and fitting-only randomness prevent a finite-sample family-wise or new-training-population guarantee. Interactions are differences of feedback effects between allocators; a positive interaction alone does not make feedback beneficial.

The three-seed completion study has 1,719 cells including references. RPL then reuses its 162 pilot cells and adds 378 to retain ten seeds, six configurations and three capacities (540 cells).

Table 2: Computational audit scope. Counts overlap where explicitly stated; human scientific approval and exact cross-machine reproduction are separate from these checks. Historical failure receipts are preserved.

Evidence	Completed chain	Boundary
Original 4,320 cells	Fit, outcome, analysis, statistics check	Original implementation; shared primitives disclosed
Claim-bearing extensions: 1,206 distinct cells	Same four computational phases; includes all 540 ten-seed RPL cells	Conditional exploratory inference; not an independent full-stack implementation
Matched RPL–acquisition crossing: 90 cells	Independent allocation/deviation/outcome checks; 18 native references reused	Three-seed descriptive effects; no new four-stage fit/statistics chain
Remaining completion-study support	Hash bindings, utility checks and recorded summaries	No whole-study four-stage upgrade of all 1,719 cells

Parameter and availability scans remain three-seed. Fit/outcome/analysis/statistics-check phases cover the original 4,320 cells and 1,206 distinct claim-bearing extension cells, including all 540 RPL cells (Table 2). Stage counts are 162 pilot RPL, 90 search, 432 SEFA/GDFS/DIME/explanation, 144 ACO and 378 additional RPL cells. The 540-cell RPL audit overlaps the pilot; remaining support cells are not upgraded. A separate matched RPL–acquisition crossing adds 90 cells with independent integer-allocation, all-legal-deviation and outcome-summary checks. Its 18 native-head reference cells are reused from the core; the new crossing has no four-stage fit/statistics-audit upgrade.

The checks reconstruct fitting roles, recorded outcomes, legal deviations and saved statistics but share prediction/loading primitives; they are not an independent full-stack implementation. An additional bit-mask oracle checks 1,188 fixture and 521,164 real-record field transitions without sharing state/query logic. Historical failures remain recorded, including the corrected ACO auditor's float32-versus-float64 distance mismatch. Human scientific approval and external numerical reproduction are separate; the latter failed its original tolerances. Appendix D indexes the complete protocols, failure receipts and audit limits.

7 Results

7.1 Recovery failures in the learned four-field class

The recovery-path prediction is observable in the saved models, rather than only in the illustrative 10/12/14 fixture. We enumerate all legal successful recovery edges in the 14 unique fits per source/seed and the original 1,000 evaluation records. An adverse partial recovery requires two hidden fields, a post-query quote above the pre-query quote and both at or above the model's honest anchor, using tolerance 10^{-10}. Table 3 reports these finite table properties separately from returned response rows.

A MUCars witness has honest, two-hidden and partially recovered log quotes 11.484823, 11.517865 and 11.546430, respectively. It hides transmission and fuel type, then recovers transmission. Thus its report component is about 0.033042 and recovery adds about 0.028564. The query raises the gain-producing quote in an actual fitted model. This witness is not a measurement of human seller behavior. Across 720 primary cells and 720,000 returned-profile rows, the saved $E + R + C = G$ identity is also reconstructed. These checks connect the recovery argument to

Source	Successful edges	Adverse partial edges	Adverse queried rows
Synthetic	2,098,600	82,939	82
JUCars	2,044,000	67,523	1,733
MUCars	2,212,560	61,404	2,829

Table 3: Recovery-path diagnostic on saved primary-capacity models. Edges use 14 unique fits per source/seed; the last column counts returned reports from all 24 cells with positive query probability and gain-increasing partial recovery. For randomized allocation these are eligible rows, not realized query counts. Each source reuses the same 1,000 records over ten fitting seeds. Counts are dependent finite diagnostics, not independent samples or deployment prevalence.

the empirical model class, but do not identify why feedback training is harmful or estimate the restricted frontier's real-data optimum.

7.2 Recovery-path regularization: improvement and costs

RPL compares $\lambda = 0, .1, .5, 1$ at $\beta = .5$ and $\lambda = 0, .5$ at $\beta = 0$, across capacities $0, .1, .3$ with the same no-feedback reset schedule. Each branch uses its same-gap zero-path control. The declared development grid predates formal evaluation; it is not independent preregistration or proof that $\lambda = 1$ was a prespecified headline. Weight .5 matches the existing gap scale; neither .5 nor 1 is claimed optimal.

Inflation improvement and uncertainty. At 10% capacity and $\lambda = 1, \beta = 0.5$, ratios of mean inflation to the matched $\lambda = 0$ control fell by 50.41%, 26.58% and 23.81% on synthetic, JUCars and MUCars. All ten fitting seeds in each source had lower inflation. The absolute paired reductions were 0.006268, 0.003306 and 0.001573 log units. The exploratory simultaneous bands over 216 metric contrasts were $[0.004644, 0.007891]$, $[0.001209, 0.005403]$ and $[0.000624, 0.002522]$, respectively (Figure 1). These conditional bands support the within-design direction without certifying finite-sample coverage or new-population generalization. Smaller weights were less uniform: at $\lambda = 0.1$, only six of ten seeds improved on each real source; at $\lambda = 0.5$, nine of ten did so. Supplement S7 retains every seed.

Loss-component boundary. The matched path/gap comparison (Figure 2) distinguishes path regularization from the existing gap term at the tested weights. Supplement S7 retains the original component ablation and three-seed recovery-edge pilot; neither is relabeled as ten-seed evidence.

Predictive and verification costs. Table 5 reports absolute outcomes. At the unit endpoint, honest MAE changed by +0.82%, −0.10% and −0.80%, while attacked MAE changed by −0.91%, −0.02% and −0.87%. Failed attempts as a fraction of all listings changed by +1.10, +0.44 and −0.12 percentage points. The synthetic honest error and failed-query increases also excluded zero in the exploratory simultaneous bands; real-source MAE changes did not. Figure 3 shows all paired seed changes, including unfavorable cases. Its means of seed ratios are distinct from the ratios of means reported above. Inflation improvement alone therefore does not determine a deployment choice. For JUCars at $\lambda = 1$, actual queries cover 10% of listings: 6.17% of all listings have failed attempts, or 61.7% of attempted queries; successful queries cover 3.83%.

Budget boundary. Figure 4 includes zero queries. The paired improvement at zero versus 10% capacity is .004276/.006268 on synthetic, .003249/.003306 on JUCars and .001527/.001573 on

MUCars. Thus zero-query improvements are 98.3% and 97.1% of the real-source 10% improvements; these ratios are descriptive, not causal shares. The paired budget interactions are .001992, .000057 and .000046. Post-hoc marginal crossed bootstrap intervals include zero for both real sources (Supplement S2). The real-source results establish an estimator improvement at the tested gap weight, without identifying an additional verification mechanism.

Recovery geometry and quote shifts. Table 4 extends the recovery-edge diagnostic to all ten fitting seeds. Positive-gain violations decrease on average, while raw quote-raising edge rates rise slightly on synthetic and JUCars. These are different targets. On MUCars, honest signed log bias rises from .029095 to .034913 and attacked bias from .035701 to .039946, despite lower MAE; their paired bias-change intervals include zero. A moving honest quote can change relative inflation. Fixed-report $E/R/C$ decompositions and endogenously reselected reports are separately reported in Supplement S3; they do not uniquely identify a causal mediator.

Table 4: Ten-seed recovery geometry at $\beta = .5$, before/after $\lambda = 1$. Rates and positive increment means use all legal concealed-field recovery edges, including zero increments; units are log price.

Source	Positive-gain violation (%)	Mean positive increment
Synthetic	15.79/5.03	.001648/.000663
JUCars	7.08/4.76	.001498/.001032
MUCars	5.27/2.60	.000743/.000393

Finite-response boundary. All 540 cells remain in Figure 5. Fifty-one cells contained nonconverged batches, covering 89 of 5,400 batch replays, with a maximum remaining unilateral regret of 0.143721 log units. At the primary unit-weight endpoint, two synthetic seeds each retained one nonconverged batch; the maximum regret there was 0.011042. Both real-source endpoint responses met the stated finite unilateral tolerance. These are outcomes of the declared finite response search, not evidence of improvement at an exact market equilibrium. Replacing primary synthetic inflation by the fixed-peer unilateral envelope changes its paired reduction from .006267693 to .006266569, a difference of 1.125×10^{-6}; this cannot explain the mean effect.

Table 5: Ten-seed RPL means at 10% capacity, $\beta = 0.5$. MAE is in source currency; failed queries are a percentage of all listings. All seeds and nonconverged responses are retained.

Source / λ	Inflation	Honest MAE	Attacked MAE	Failed (%)
synthetic / 0	0.01243	3576.06	3635.45	2.51
synthetic / 0.5	0.00707	3595.31	3600.02	3.49
synthetic / 1	0.00616	3605.35	3602.26	3.61
jucars / 0	0.01244	6966.89	6957.18	5.73
jucars / 0.5	0.01077	6948.32	6941.48	5.35
jucars / 1	0.00913	6959.75	6956.01	6.17
mucars / 0	0.00661	27155.74	27306.07	0.43
mucars / 0.5	0.00566	26983.89	27114.40	0.39
mucars / 1	0.00503	26938.93	27069.81	0.31

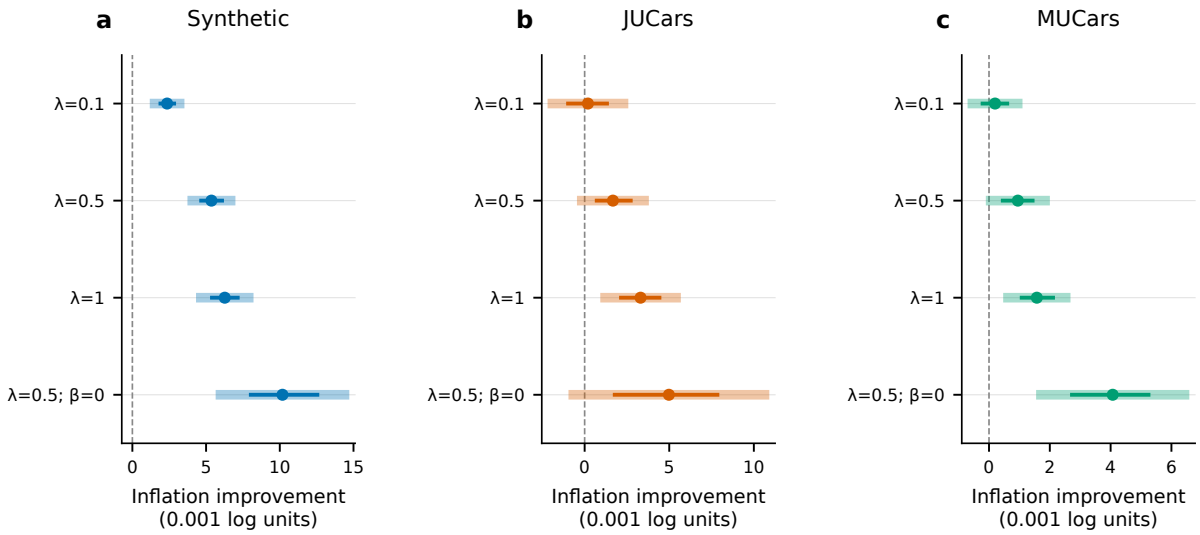


Figure 1: Ten-seed paired inflation improvement at 10% capacity. Positive values favor RPL. Each branch uses its same-gap zero-path control. Thin segments are marginal 95% intervals; pale segments are exploratory simultaneous bands over 216 metric contrasts. Ten fitting seeds share fixed training/evaluation populations and ten batches. Bands do not establish finite-sample family-wise coverage.

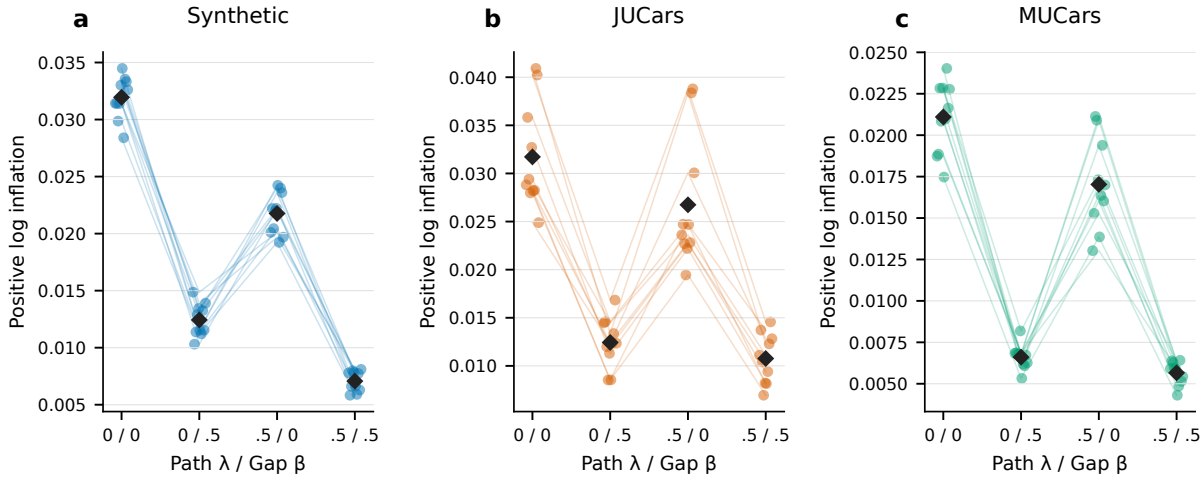


Figure 2: Matched path/gap loss-component comparison at 10% capacity. Each line connects one fitting seed across four configurations; dots retain all observations and black diamonds show means. The axis lists path weight/gap weight. This tests the path term against matched zero-path controls with and without the gap term.

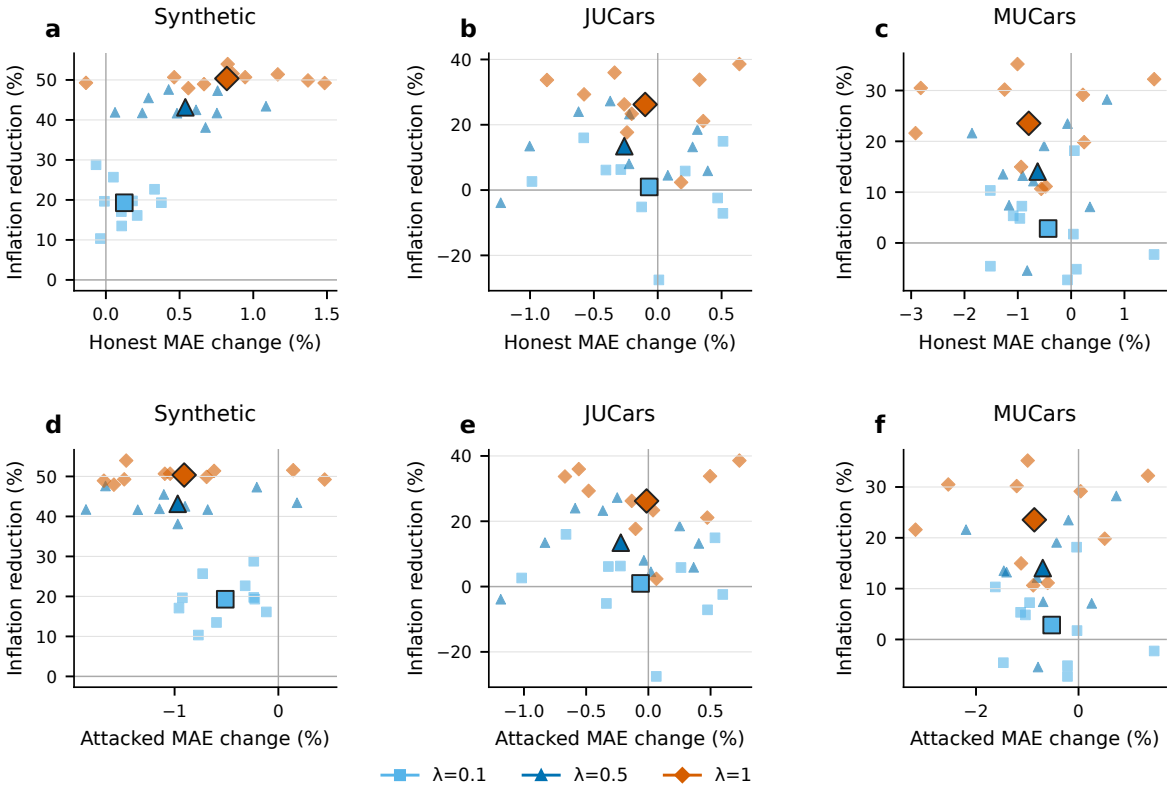


Figure 3: Paired seed trade-offs at 10% capacity and $\beta = 0.5$. Small markers retain all ten fitting seeds per source and positive path weight; outlined markers are their means. Horizontal axes show honest or attacked MAE changes, while vertical axes show inflation reductions. Upper-left is favorable. Each point uses its same-seed control, so these mean seed ratios differ from the headline ratios of means. No confidence region is claimed.

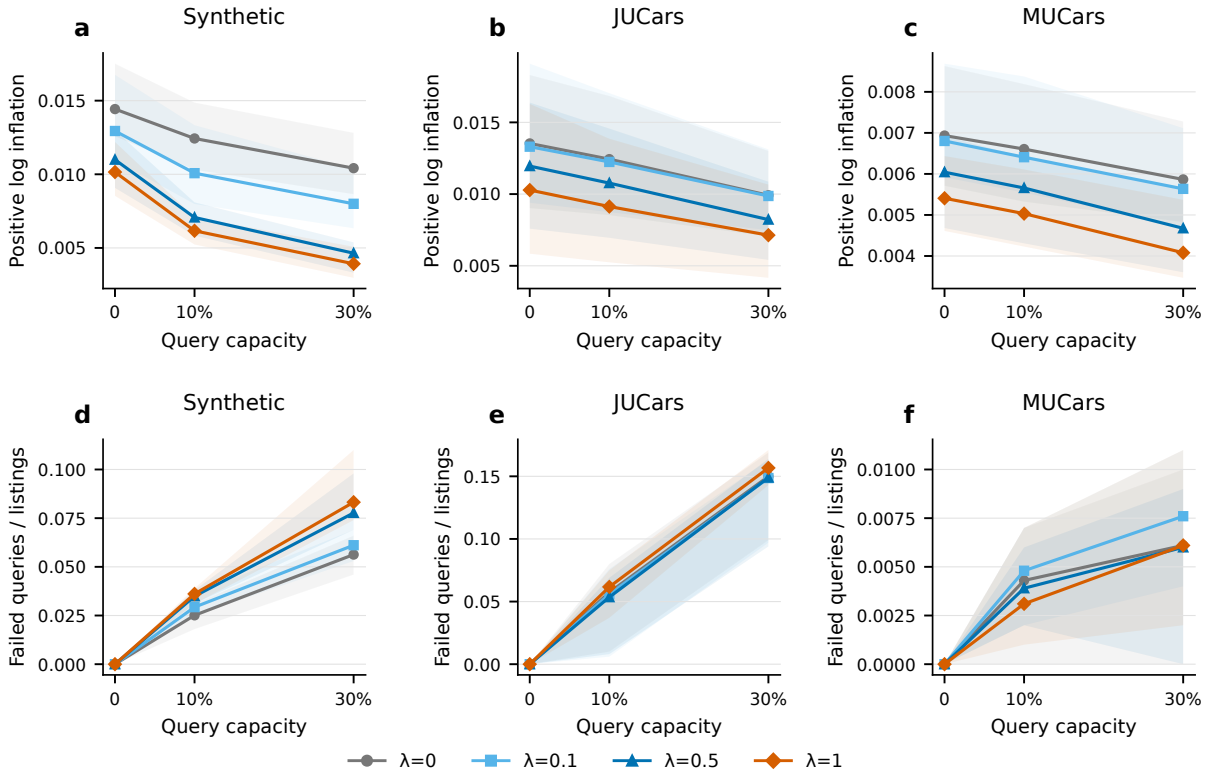


Figure 4: Budget transfer and verification costs at $\beta = 0.5$. The measured capacities are 0, 10 and 30%; straight segments connect evaluated settings only. Lines show ten-seed means and shading the full seed range, not a confidence interval. Failed queries use all listings as the denominator. Zero-query differences are estimator effects.

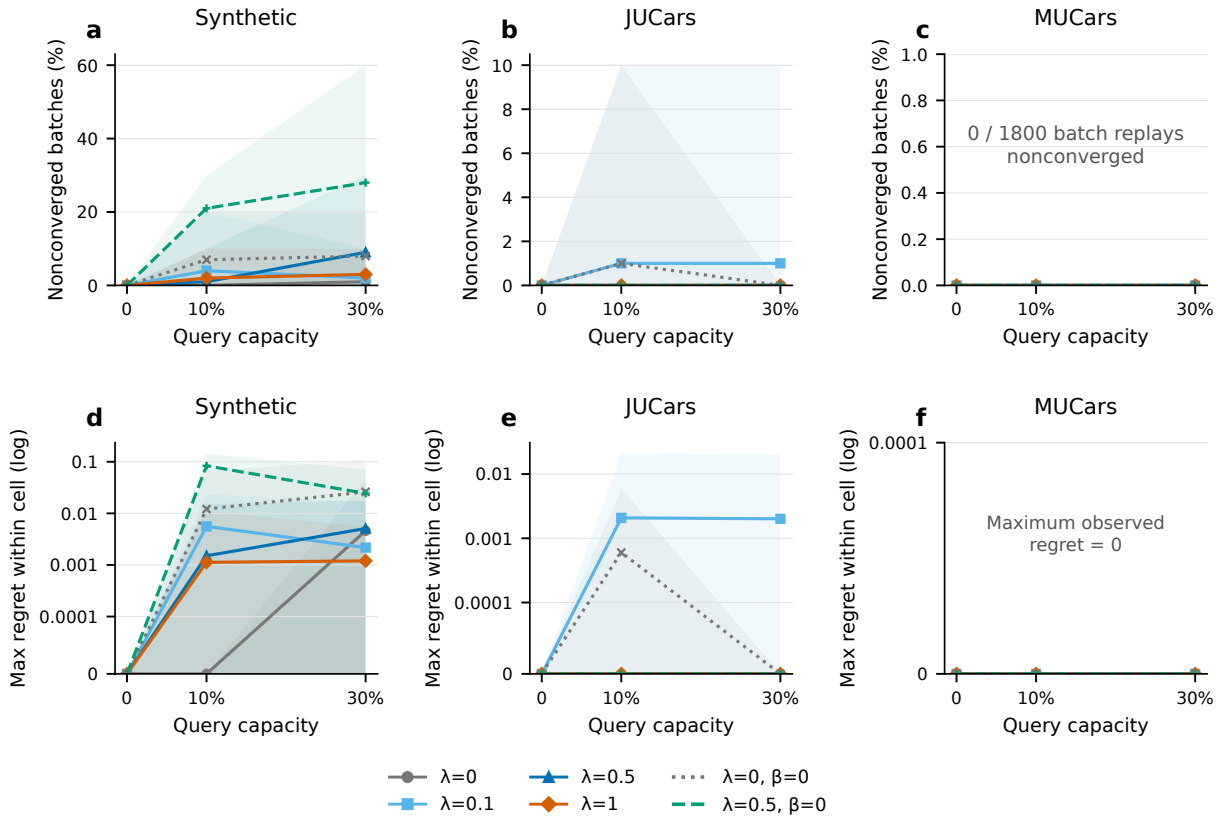


Figure 5: Convergence boundary across all 540 RPL cells and six configurations. Top panels show batch nonconvergence rates; bottom panels show the maximum unilateral regret within each cell on a symmetric-log axis. Curves are ten-seed means, bands the full seed range. The mean of cell maxima is not the worst observed regret. All 51 cells containing nonconverged batches are retained.

7.3 Matched RPL and five recent acquisition families

The direct comparison crosses $\lambda = 0, 1$ with ACO, SEFA, GDFS, DIME and explanation-ranked AFA at $\beta = .5$ and 10% capacity: three sources and the same three fitting seeds give 90 cells. Both predictors use the same no-feedback training schedule and cohorts. The four neural acquisition tables remain frozen; ACO's training-bank losses are recomputed under each predictor, with $k = 25$ fixed. No policy is selected or refitted on evaluation outcomes. These are task adaptations, with code/observation/selection boundaries in Appendix C, not official end-to-end reproductions of the original papers.

Does RPL help each acquisition family? Table 6 shows both predictor settings and matched costs. Adding RPL lowers inflation in all 45 source–family–seed pairs. Ratios of family means fall by 32.97–48.00% on synthetic, 30.56–32.70% on JUCars and 18.97–25.55% on MUCars. Within each source, even the largest RPL-on family mean is below the smallest RPL-off family mean. Thus the inflation gain persists across these acquisition choices in the fixed design.

Does stacking improve on the native RPL pipeline? Eighteen existing matched cells retain the original suppression head (*native head*); it still performs acquisition. At $\lambda = 1$, SEFA has the lowest observed inflation mean on synthetic and GDFS on both real sources, respectively 17.07%, 4.76% and 22.47% below native-head RPL. These retrospective minima are not selected deployment defaults. Stacking is not uniformly better than native-head RPL: synthetic GDFS gives 0.00907 versus the native head's 0.00639. The best complete pipeline therefore depends on the acquisition family and objective.

Costs and evidence boundary. Across the matched three-seed predictors, honest MAE changes by $+0.99\%, -0.19\%, -0.91\%$ on synthetic/JUCars/MUCars, independently of acquisition. All five-family attacked-MAE means decrease, while failed-query changes range from -0.07 to $+1.23$ percentage points per listing. All 18 nonconverged batches among the 900 crossing batches remain included; maximum remaining regret is 0.14916 log units. This is three-seed development evidence on shared cohorts; seed ranges are not confidence intervals or the ten-seed study. Appendix C gives absolute error and failure means; source CSVs retain every seed and convergence value.

7.4 Controlled components and a search-strength intervention

The complete suppression-score/feedback/gap factorial retained all eight combinations, including removal of two or all three switched components (Supplement S7). These switches are experimental components, not three independently established innovations. Gap removal substantially increased mean primary inflation on the real sources, whereas the full three-component configuration was not uniformly best.

The original scoring network was refreshed each epoch, rather than frozen throughout training. We crossed epoch versus minibatch score refresh with weak (two sweeps, one start) versus strong (20 sweeps, four starts) training-response search. Both retained two six-epoch rounds and matched no-feedback references. Figure 6 shows the audited paired improvements and conditional intervals at 10% capacity, retaining three fitting seeds. On synthetic, epoch-weak inflation was 0.02075, epoch-strong 0.01279, batch-strong 0.01269, and no-feedback 0.01303. Thus strengthening search substantially reduced the weak implementation's damage, and the strong means were slightly below the no-feedback reference. Strong JUCars means were also slightly lower than no-feedback, whereas all four MUCars feedback means remained higher. Changing score-refresh frequency alone was not

Table 6: Matched RPL–acquisition comparison at 10% capacity and $\beta = .5$. Each entry averages the same three fitting seeds. I is positive log inflation; ΔA is the attacked-MAE change (percent), and ΔF the failed-query change (percentage points per listing), from $\lambda = 0$ to 1. Native head rows reuse 18 existing cells; the other rows are the new 90-cell 2×5 crossing. Negative changes favor the indicated metric.

Source	Acquisition	$I_0 \downarrow$	$I_1 \downarrow$	ΔI (%)	ΔA (%)	ΔF (pp)
Synthetic	Native head	0.01303	0.00639	-50.93	-0.91	+0.63
	ACO	0.00988	0.00537	-45.64	-0.16	+0.03
	SEFA	0.01019	0.00530	-48.00	-0.17	+0.13
	GDFS	0.01353	0.00907	-32.97	-0.75	-0.07
	DIME	0.01306	0.00782	-40.11	-0.67	+1.23
	Expl.-AFA	0.01388	0.00903	-34.89	-0.79	+0.10
JUCars	Native head	0.01380	0.00922	-33.22	-0.17	+1.47
	ACO	0.01401	0.00952	-32.06	-0.19	+0.57
	SEFA	0.01338	0.00929	-30.56	-0.20	-0.07
	GDFS	0.01290	0.00878	-31.95	-0.16	+0.87
	DIME	0.01396	0.00939	-32.70	-0.16	+0.67
	Expl.-AFA	0.01294	0.00880	-32.00	-0.22	+0.73
MUCars	Native head	0.00666	0.00541	-18.72	-1.07	+0.00
	ACO	0.00575	0.00460	-20.11	-0.98	+0.30
	SEFA	0.00549	0.00425	-22.51	-1.10	+0.23
	GDFS	0.00563	0.00419	-25.55	-1.05	+0.30
	DIME	0.00641	0.00519	-18.97	-1.07	+0.20
	Expl.-AFA	0.00557	0.00432	-22.38	-1.03	+0.17

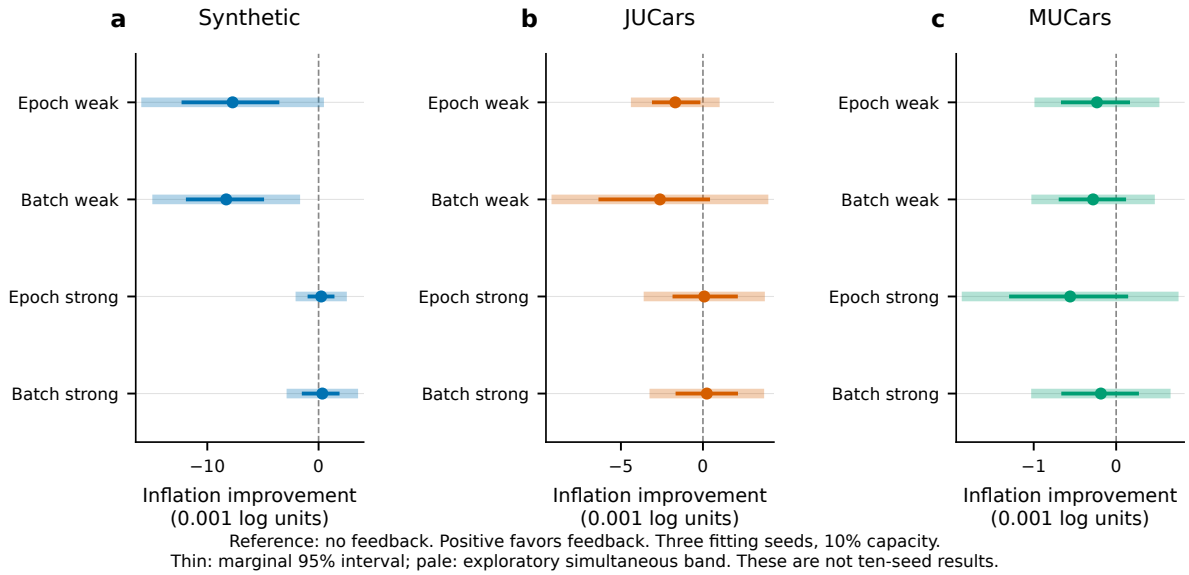


Figure 6: Search-strength and score-refresh interventions relative to matched no-feedback controls, at 10% capacity. Points are mean reference-minus-candidate inflation differences; positive values favor feedback. Thin segments are marginal 95% crossed-bootstrap intervals and pale segments are exploratory simultaneous bands. This intervention retains three fitting seeds and ten shared batches; it is separate from the ten-seed RPL extension.

a consistent remedy. These are conditional mean comparisons, not proof of a uniquely identified cause or population-wide feedback benefit.

7.5 Historical feedback and allocation controls

The original factorial had adverse mean inflation directions in all 36 primary feedback contrasts. Under post-hoc exploratory simultaneous families, 31 adverse intervals remained for the inflation-only family, 27 for three co-primary outcomes, and 20 for all 1,536 primary metric contrasts. No family established that every feedback contrast was harmful. These results describe the original weak-search implementation; the stronger-search intervention above changes that interpretation. Appendix D indexes all original intervals, interactions and counts.

The strong-control comparisons also restrict claims for the original SAV candidate. Ordinary-benefit and acquisition-conditioned controls had lower inflation in synthetic and JUCars comparisons, and the original ten-seed SAV comparison did not show a universal advantage over static robust valuation. These original-protocol comparisons are distinct from the new same-gap RPL contrasts. Likewise, score-minimax allocation passed the descriptive joint gate in only two of eighteen primary contrasts, while ten worsened inflation. An advantageous interaction or a more complex allocator therefore does not replace a matched no-feedback or simple-acquisition control. Complete three-source evidence is retained in Appendix D.

7.6 History-cap outcomes and deployment costs

The history cap's fixed-peer theorem does not imply a better joint outcome after sellers recompute their reports. The 432-cell recomputation kept the original response-search budget, public scores and rule weights. Its multiobjective result is consequential: attacked MAE increased in 94 cells

and failed-query rates in 145, despite unchanged honest MAE. Figure 7 shows those costs alongside inflation reductions. Complete per-rule counts and conditional intervals are indexed in Appendix D.

Thirty-three capped cells retained at least one nonconverged batch and maximum remaining regret was 0.204423 log units. No capped result is therefore described as a certified market equilibrium. The descriptive 10% inflation / 5% honest-error gate does not constrain attacked error or failed queries; threshold sensitivity cannot turn a single-metric improvement into Pareto dominance. These costs also motivate the final-record alternative and its natural-missingness boundary in Section 5.

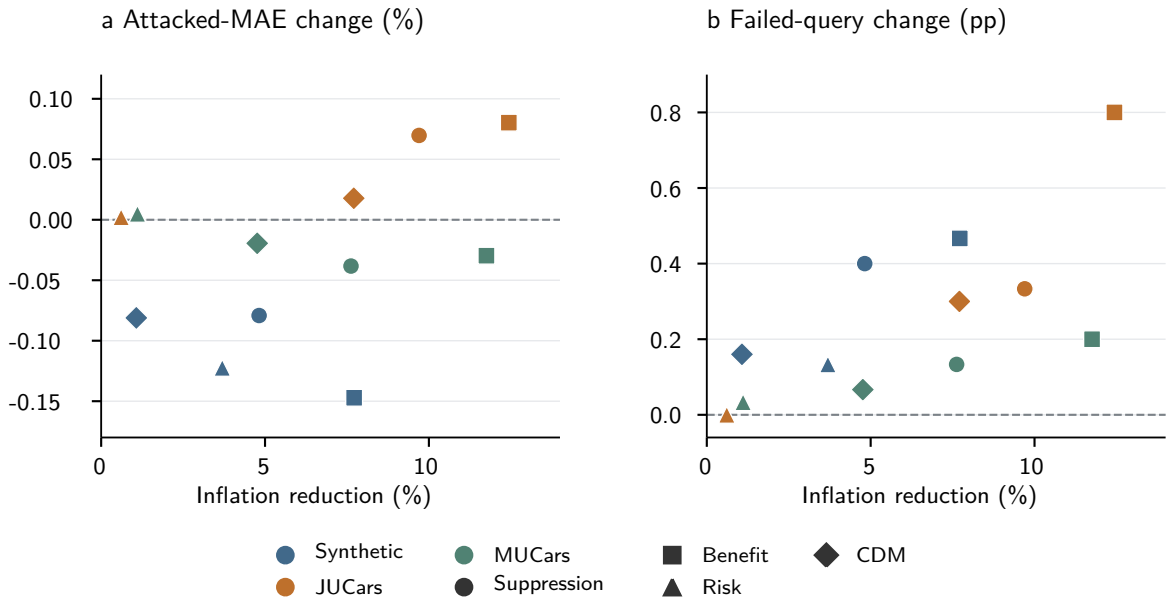


Figure 7: Inflation reductions under the history cap coexist with predictive-error and query-failure costs. Both panels show the same twelve primary robust-estimator source/rule means at 10% capacity, averaging three fixed fitting seeds on 1,000 shared listings per source. Colors denote sources and shapes denote query rules. Positive inflation reduction is favorable; positive attacked-MAE change (a) and failed-query change (b) are unfavorable. Error changes are within-source percentages; failure changes are percentage points per listing. These are descriptive means, without confidence intervals; nonconverged responses remain included. The full numerical comparison is indexed in Appendix D.

7.7 Applicability beyond the zero-cost capped design

Positive concealment costs changed some rankings through deterrence. In the synthetic top-B subset, stipulated costs near 0.02 sharply reduced concealment and moved outcomes toward ordinary valuation; feedback no longer worsened the mean inflation in the high-cost specifications. Yet over 94% of attempted queries failed in the uniform-cost subset, residual concealment remained positive, and the real-source directions stayed adverse. This supports a conditional regime change rather than a general feedback remedy. The full 108-cell cost design and net-deviation bounds remain in Appendices D and B.

A separate frozen-original-model check removed the real-source evaluation caps, covering all

1,374 JUCars and 10,672 MUCars validation records. All 24 feedback point directions remained adverse. The two MUCars masking/suppression intervals still crossed zero, with fitting-seed variability dominating the uncapped conditional variance. The cohort and record-count weighting both changed, so this is not a pure sample-size experiment or evidence of zero effect. It does not extend the RPL evaluation population: the RPL ten-seed study retains its fixed 1,000 listings per source. Complete uncapped counts, weighting and variance decompositions are in Appendices D and B.

8 Discussion and conclusion

The information frontier limits guarantees under indistinguishable reports; observed recovery paths diagnose a separate failure of fitted predictors. RPL links that diagnosis to learning and lowers inflation at the tested unit-weight endpoint, without private deployment inputs. Its zero-query component accounts descriptively for nearly all real-source improvement. The evidence supports recovery-graph-inspired regularization; it does not identify recovery as the main causal source of that improvement. Synthetic honest-error and failed-query increases, together with finite response regret, preclude a universal dominance claim. The matched crossing supports RPL as a complement to acquisition at the tested weights and cohorts. Native-head counterexamples and error/failure costs keep the deployment decision specific to the complete predictor–policy pair.

Stronger-search interventions restrict the earlier negative feedback result to its weak-search implementation; they do not identify a unique causal bug or establish feedback benefit across sources. The history cap similarly has a useful fixed-peer guarantee but can worsen joint outcomes and raises same-record consistency issues. Decisions should therefore use the joint incentive, prediction and verification outcomes rather than inflation alone.

The cohorts are retrospective advertised-price samples. Concealment, costs and response dynamics are modeled, not observed seller behavior; fitting seeds do not sample new training populations. Shared audit primitives and nonconverged profiles constrain computational and equilibrium interpretations. An independently operated Mac completed six synthetic/seed-13 training cells: inputs and tree backbones matched, but residual parameters and outcomes failed the original numerical tolerances. Later diagnostics locate the first recorded difference in output-layer gradients; unrecorded hidden activations prevent causal attribution to a particular primitive. One-seed directions agreed, which is not numerical equivalence. At 10% capacity, the local/external paired inflation reductions are 53.97/57.54%, but maximum residual regret for $\lambda = 1$ changes from .000204 to .021973. Supplement S5 reports all six cells under the original failed tolerance verdict. An isolated float64 candidate has not replaced the formal experiments. Actual seller data and independently controlled numerical environments remain necessary transportability tests.

Statements and declarations

Data and code. Datasets are available under their versioned CC BY 4.0 records [1, 2]. The public research repository contains project-authored MIT code, protocols, aggregate results, figure data, resampling tensors and receipts; third-party software/data retain their own licenses. Large weights and per-listing artifacts remain on the research machine. Software and aggregate evidence through source commit `89ad56e` are archived at DOI 10.5281/zenodo.23447567. The later review diagnostics and stronger-gap comparison are subsequent repository evidence, not additions to that immutable ZIP. Supplement S6 maps versions and artifacts.

AI assistance and author responsibility. OpenAI Codex assisted implementation, orchestration, analysis, literature checks, data-derived figures and drafting. Figures use saved empirical results, not invented images. Authors retain responsibility for interpretation and approval. Yuchen Tang drafted the manuscript. Fengqin You provided supervision as advisor, and Junxiang Gu provided guidance on manuscript writing.

A Contextual policy-library witnesses

Take $n > B \geq 1$, a core C of size B, and public active contexts $j \in \{B+1, \ldots, n\}$. Gaps in linear toy units are 2 for core items, 1 for j, and $\epsilon = 1/(B+2)$ for other items, using honest-exact missingness-alias worlds. The library includes every context-insensitive priority order and exact-gap risk ordering, but shares one mixing vector. Each priority chooses a fixed B-subset; risk always chooses the core.

Proposition 3 (Global-mixture separation)**.** *The optimal values are*

$$T_{\text{global}} = \frac{2(n-B)}{2n-B}, \qquad T_{\text{context}} = \frac{2}{B+2}.$$

For $n = B^2 + B$, their ratio $B(B+2)/(2B+1)$ is unbounded in B.

Proof. A global value t requires $2(1-p_i) \leq t$ on core items and $1-p_i \leq t$ on every potentially active item. Summing yields $B \geq n - t(B/2 + n - B)$. Core marginals $1 - T_{\text{global}}/2$ and other marginals $1 - T_{\text{global}}$ attain equality through fixed-priority lotteries. For one context, $B \geq B+1-t(B/2+1)$; core marginals $(B+1)/(B+2)$, active marginal $B/(B+2)$ and others zero attain its bound, with $\epsilon < T_{\text{context}}$. Proposition 1 implements both. □

A stronger fixed library closes the gap: one rule queries the core and each of B others replaces one core item by the active item. Fixed weights $2/(B+2)$ on the core rule and $1/(B+2)$ on each replacement yield exactly these contextual marginals. Thus the separation concerns rule information, not dynamic weights or the intrinsic necessity of an LP solver.

Exact witnesses cover 26 (n, B) cases, 257,274 context–item losses and 15,922 context–rule allocations; sixteen small cases match independent complete-subset LPs. At $B = 20, n = 420$, the ratio is 440/41. Freezing the active item or using uniform rule weights fails the optimum check. These constructed results neither explain real-source CDM loss nor guarantee unseen-context behavior.

B Mechanism and uncertainty boundaries

Conditional uncertainty. For paired differences Y with S fitting seeds, independent bootstrap seed weights U and normalized batch weights V give $U^\top Y V$. Set $m = \mathbb{E}[V]$, $C = \text{Cov}(V)$, $r = Ym - \mathbf{1}\,\text{mean}(Ym)$, $z = \text{mean}_s\, Y$ and $R = Y - \mathbf{1}z^\top - r\mathbf{1}^\top$. The exact conditional variance is

$$\|r\|^2/S^2 + z^\top C z + S^{-2} \sum_s R_s^\top C R_s.$$

The seed, batch and interaction terms were independently checked on balanced and unequal 2×2 cases. All 48 real-source decompositions remain in the repository. Capped analysis uses ten equal batches; uncapped MUCars uses 107 record-weighted batches including a 72-record tail. The outcome matrix and weighting both change: batch terms fall by 5.73/8.04 for top-B/minimax, while seed terms change by $-13.20\%/-9.43\%$. Within all 107 batches, changing only equal-batch to record-count weights changes the seed term by $-0.16\%/+0.11\%$. Thus neither expanded coverage nor its variance shares are a pure sample-size experiment. Over 98% fitting-related shares describe the uncapped design only (capped: 87–91%); no equivalence or zero-effect claim follows from intervals crossing zero.

Positive-cost mechanism. The 108-cell post-hoc subset uses robust initialization/suppression scoring, three seeds, both feedback settings and allocators, with cost .005, .02 or $.02h_i$, $h_i \in [.5, 1.5)$. Responses are recomputed, not adjusted by subtracting costs afterward. For cost $C_i = c_i k_i$, positive quote gain G_i, positive best-legal-utility gain N_i and remaining regret ρ_i, $C_i - \rho_i \leq G_i - N_i \leq C_i$. These bounds, not division by mean heterogeneous cost, identify concealment ranges. Twenty nonconverged batches remain. Synthetic top-B cost-.02 hidden-field means are bounded by $[.130, .165]/[.130, .181]$ without/with feedback, versus .996/.957 at zero cost; net-deviation envelopes fall 77.5%/81.1%. Failed queries reach 9.47%/9.40% per listing (at least 94.7%/94.0% of attempts). High costs move the subset toward ordinary valuation and feedback no longer worsens mean inflation, but residual concealment stays positive. Uniform-.02 benefit appears in only one seed, heterogeneous-cost benefit in all three; real-source directions remain adverse. This supports deterrence alongside a local ranking reversal, not causally identified improvement of feedback quality.

Final-record envelope. Three robust/no-feedback fits per real source enumerate categorical transmission/fuel/condition completions from training vocabularies, with mileage observed and fixed. Unsupported categories or missing mileage abstain. Current records are included; singleton domains leave quotes unchanged. Of 999 covered MUCars records, 971 have singleton completions and zero cost; the 28 naturally incomplete records concentrate its error increase. Of 801 covered JUCars records, 800 are singletons and all 801 retain the same quote. The largest individual MUCars quotation reduction is 25.5% of that record's original predicted quote. Quote discounts and absolute-error costs have different denominators. Thus the restricted envelope is costless on complete covered records but can be expensive with natural missingness. Table 7 retains all pilot ranges; these are not confidence intervals or a tested four-field replacement.

Source	Support	Records	Honest-MAE change	Mean quote reduction
JUCars	Covered	801	0	0
JUCars	Missing category	1	0	0
MUCars	Covered	999	10.49–23.87	200.21–240.51
MUCars	Missing category	28	374.24–851.51	7,143.33–8,581.21

Table 7: Final-record categorical-envelope pilot. Entries are per-source three-seed ranges, not uncertainty intervals; positive MAE changes are worse. Price quantities use source currency units. Missing-category rows are subsets of covered rows, not additional observations. JUCars abstains on 199 records, MUCars on one, all for missing mileage.

C Recent-method adaptations and numerical comparison

The matched comparison in Table 8 uses the same three sources, fitting seeds 13/42/2026, evaluation records, batch ordering, gap weight .5, no-feedback schedule and 10% capacity for both RPL settings. Its 90 cells add no fits: four neural acquisition tables reuse their frozen training-only choices, while ACO reconstructs all sixteen guidance-bank observation losses under each predictor. The native suppression heads reuse 18 previously audited cells with identical predictor/evaluation hashes and listing order. Their three-seed means are distinct from the ten-seed means in Table 5. Per-seed outcomes, full ranges and convergence values are at `notes/rpl-paper-integration-v1/`.

All adapters share a one-query hard cap, public nonquery context and unrecoverable natural absence. The four classification adapters train on four price-quantile classes; using their scores

as cross-listing priorities has no cardinal-comparability theorem. Concealed-mileage perturbations leave the relevant public-state scores unchanged. The earlier two fixed-backend, six-budget study (540 cells plus 36 ACO sensitivities) remains in `outputs/recent-five-analysis-v1/`; its budget curves are preserved at `paper/figures/postcompute-v1/recent_budget_curves.pdf`. The present table answers the matched RPL/stacking question instead.

SEFA uses official MIT code (commit `3af8da2`), configurations 1/5/9 and 120 epochs/trial. GDFS/DIME use that repository's comparison implementations: GDFS retains relaxed one-hot selection and temperature annealing; DIME retains entropy-scaled values and epsilon-greedy choice. They use 80 pretraining epochs and 15 per annealing stage. They start from legal public states, train at most two recoveries and do not reproduce the original learning-rate scheduler. Explanation-ranked AFA uses official network classes (`d08f39a`): MLP and three-layer/four-head GPT, hidden width 128, embedding width 64. A training-only 250-tree classifier supplies summed field/missing-indicator TreeSHAP ranks. Its masked prediction, ranking and trajectory stages use 250, 200 and 16 epochs, respectively, with two recoveries and a fixed initial marker; learning rates .001/.0005/.0003 form the local grid. All four select three configurations with a training-only 80:20 split, then refit all training records. SEFA selects masked validation NLL/epoch; the other three select mean pre/post-query NLL and retain full schedules. Selection objectives therefore differ. These are documented task adaptations, not official end-to-end reproductions.

ACO-1 is a clean-room regression adaptation. A training-only neighbor bank estimates loss changes for sixteen observation patterns, using observed query coordinates and fixed $k = 25$ neighbors (fixed sensitivities 10/25/50). Unrecoverable bank values give zero benefit. Classic risk/benefit heads, static robust, CDM and the adapted support-margin control remain comparators. The support-margin categorical fallback rates are 75.3%, 87.3% and 65.6% on synthetic/JUCars/MUCars; its 94,527-case enumeration check does not restore the original guarantee on fallback rows. These support-margin comparisons do not isolate acquisition effects.

Shared honest errors. For $\lambda = 0$ and 1, honest MAE is respectively 3566.27 and 3601.64 (synthetic), 7084.30 and 7070.78 (JUCars), and 27154.31 and 26906.52 (MUCars). Each pair is shared by all six acquisition choices. Table 8 gives their remaining absolute outcomes.

Table 8: Absolute matched outcomes at 10% capacity, $\beta = .5$, three seeds. Each slash separates $\lambda = 0/1$. MAE is in source currency; failed queries are percent of all listings. All nonconverged profiles remain included.

Source	Acquisition	Inflation 0/1	Attacked MAE 0/1	Failed (%) 0/1
synthetic	Native head	0.01303/0.00639	3642.04/3608.84	2.87/3.50
synthetic	ACO	0.00988/0.00537	3608.08/3602.35	3.47/3.50
synthetic	SEFA	0.01019/0.00530	3601.58/3595.37	3.40/3.53
synthetic	GDFS	0.01353/0.00907	3650.50/3622.99	3.27/3.20
synthetic	DIME	0.01306/0.00782	3636.75/3612.32	2.27/3.50
synthetic	Expl.-AFA	0.01388/0.00903	3655.20/3626.37	3.23/3.33
jucars	Native head	0.01380/0.00922	7078.99/7066.68	4.37/5.83
jucars	ACO	0.01401/0.00952	7079.86/7066.35	2.23/2.80
jucars	SEFA	0.01338/0.00929	7080.43/7066.00	3.83/3.77
jucars	GDFS	0.01290/0.00878	7076.49/7065.10	4.73/5.60
jucars	DIME	0.01396/0.00939	7075.43/7064.21	4.07/4.73

Source	Acquisition	Inflation 0/1	Attacked MAE 0/1	Failed (%) 0/1
jucars	Expl.-AFA	0.01294/0.00880	7081.03/7065.39	4.20/4.93
mucars	Native head	0.00666/0.00541	27363.19/27070.79	0.33/0.33
mucars	ACO	0.00575/0.00460	27335.00/27068.14	0.30/0.60
mucars	SEFA	0.00549/0.00425	27319.44/27019.15	0.27/0.50
mucars	GDFS	0.00563/0.00419	27345.27/27057.07	0.57/0.87
mucars	DIME	0.00641/0.00519	27349.46/27056.62	0.60/0.80
mucars	Expl.-AFA	0.00557/0.00432	27317.58/27035.70	0.03/0.20

D Historical controls and reproducibility index

The public repository preserves the full pre-compression manuscript at commit `f5da885`, including the original 17 figures and 18 tables. Current figure source data, all fitted comparisons and failed receipts remain available. This compact appendix summarizes secondary controls rather than duplicating their full tables; no experiment, negative outcome or nonconverged row was excluded.

Original comparisons. In ten-seed original-protocol comparisons at 10% capacity, comparator-minus-SAV inflation differences for static robust are −.009091 (synthetic), −.002906 (JUCars) and −.000829 (MUCars). The synthetic interval excludes zero; both real-source intervals cross it. These 620 cells/source are not the matched factorial. In separately recomputed common-response controls, ordinary benefit and ACO beat SAV on synthetic/JUCars; some larger-cap MUCars comparisons favor SAV. No evaluation-best tuning cap replaces the declared role. CDM's finite rows/library and adapted support-margin fallbacks also preclude a universal original-method advantage.

Allocation and history cap. Among eighteen primary allocator contrasts, score-minimax worsens inflation in ten; only two pass the descriptive joint gate. Gate cutoffs of 10% inflation reduction and 5% honest-error increase are retrospective design thresholds, not utility-identified requirements. Continuous effects and threshold grids remain in the record, including JUCars' unrounded 9.985% case. The fixed-profile cap check covers 108 cases and 5,671,620 legal rows; all honest quotes are unchanged and envelopes weakly decrease, as predicted. Risk's primary improvement is zero. After recomputing seller responses, 432 cells contain 94 attacked-MAE and 145 failed-query increases; 33 cells retain nonconverged batches, with maximum regret .204423. The mechanically nonincreasing fixed-profile cap is not a new empirical dominance finding.

Expanded cohorts. Removing both original-model real-source caps evaluates all 1,374/10,672 JUCars/MUCars records: 480 cells, 29,040 batches, 2,891,040 rows and 31,269,840 legal deviations. First-ten-batch records and receipts are reused byte-for-byte; new batches use no retraining. Final batch sizes 74/72 use budget 7 at 10%. All returned real-source batches meet the finite unilateral tolerance, with recorded maximum regret zero. All 24 feedback directions stay adverse, 12 JUCars and 10 MUCars unadjusted intervals exclude zero; the two MUCars masking/suppression intervals still cross zero. These checks concern frozen original models, not an uncapped RPL population.

Audit coverage and surviving failures. Original fit checks reconstruct 2,068,360 held-out rows and refit 420 heads; outcome checks cover 4.32 million rows and 45,372,960 legal deviations;

Table 9: Compact evidence map. Paths are relative to the public repository.

Evidence	Public location / scope
Original factorial	`notes/factorial-final/`; 4,320-cell chain
Core extensions	`notes/core-evidence-v2/completed-stages/`; 1,206 unique cells
RPL ten-seed effects	`notes/ten-seed-visual-review-v1/`; all seeds and figure CSVs
RPL–acquisition comparison	`notes/rpl-paper-integration-v1/`; 90 crossed + 18 reused cells
Expanded real cohorts	`notes/revision-review-v2/evaluation/`; 480 cells
Cost/envelope diagnostics	`notes/revision-mechanisms-v3/`; source-bound checks
Strict external mismatch	`notes/external-reproduction-synthetic-v1/`; failures retained
Original and revised figure assets	`paper/figures/`; vector and raster sources

statistics reconstruction checks 9,216 metric contrasts. The separate state oracle rejects four injected transition/materialization faults but does not reimplement datasets, predictors or optimizers. The new 1,206-cell chain uses shared primitives too. The initial ACO auditor subtracted float32 bank/query arrays, while production promotes both to float64 before distances. The separate corrected auditor reproduces all 144 ACO cells without changing production, results or tolerance; the failed receipt is retained. Earlier argmax-identity failures arose from ties (largest payoff gap 6.94×10^{-18}); saved original winners remain. Neither mutation rejection nor computational agreement is independent scientific approval.

References

[1] N. Thalji. JUCars-2024: A Dataset of Used Car Listings in Jordan. Mendeley Data, version 2, 2026. doi:10.17632/ddcz486x5t.2.

[2] E. Tabarnoust, M. Mghari and Y. Zaz. Moroccan Used Cars Dataset (MUCars – 2024). Mendeley Data, version 2, 2024. doi:10.17632/vjrbcb2rrt.2.

[3] J. Letchford and V. Conitzer. *Solving Security Games on Graphs via Marginal Probabilities*. AAAI, 2013, pp. 591–597. `https://doi.org/10.1609/aaai.v27i1.8688`.

[4] A. Estornell, S. Das and Y. Vorobeychik. *Incentivizing Truthfulness Through Audits in Strategic Classification*. Proceedings of the AAAI Conference on Artificial Intelligence, 35(6):5347–5354, 2021. `https://doi.org/10.1609/aaai.v35i6.16674`.

[5] X. Liu, X. Han, N. Zhang and Q. Liu. *Certified Monotonic Neural Networks*. Advances in Neural Information Processing Systems, 2020. `https://arxiv.org/abs/2011.10219`.

[6] A. K. Krishnaswamy, H. Li, D. Rein, H. Zhang and V. Conitzer. *Classification with Strategically Withheld Data*. Proceedings of the AAAI Conference on Artificial Intelligence, 35(6):5514–5522, 2021. `https://doi.org/10.1609/aaai.v35i6.16694`.

[7] H. Zhang, Y. Cheng and V. Conitzer. *Automated Mechanism Design for Classification with Partial Verification*. Proceedings of the AAAI Conference on Artificial Intelligence, 35(6):5789–5796, 2021. `https://doi.org/10.1609/aaai.v35i6.16725`.

[8] P. G. Sessa, I. Bogunovic, A. Krause and M. Kamgarpour. *Contextual Games: Multi-Agent Learning with Side Information*. NeurIPS, 2020. `https://papers.nips.cc/paper/2020/file/f9afa97535cf7c8789a1c50a2cd83787-Paper.pdf`.

[9] R. Ebrahimi and M. Franceschetti. *Strategic Classification Has a Missing Lever: Audit Risk*. arXiv:2609.22534v1, 18 September 2026, preprint. `https://arxiv.org/abs/2609.22534`.

[10] J. Zhang, X. Liu and X.-B. Li. *Predictive Analytics with Strategically Missing Data*. INFORMS Journal on Computing, 32(4):1143–1156, 2020. `https://doi.org/10.1287/ijoc.2019.0947`.

[11] J. Perdomo, T. Zrnic, C. Mendler-Dünner and M. Hardt. *Performative Prediction*. ICML, PMLR 119:7599–7609, 2020. `https://proceedings.mlr.press/v119/perdomo20a.html`.

[12] S. Levanon and N. Rosenfeld. *Strategic Classification Made Practical*. ICML, PMLR 139:6243–6253, 2021. `https://proceedings.mlr.press/v139/levanon21a.html`.

[13] M. Valancius, M. Lennon and J. Oliva. Acquisition Conditioned Oracle for Nongreedy Active Feature Acquisition. ICML, PMLR 235:48957–48975, 2024. https://proceedings.mlr.press/v235/valancius24a.html.

[14] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush and A. Gulin. CatBoost: Unbiased Boosting with Categorical Features. Advances in Neural Information Processing Systems 31, 2018. https://proceedings.neurips.cc/paper_files/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html.

[15] P. Geurts, D. Ernst and L. Wehenkel. Extremely Randomized Trees. Machine Learning 63:3–42, 2006. https://doi.org/10.1007/s10994-006-6226-1.

[16] A. B. Owen. The Pigeonhole Bootstrap. The Annals of Applied Statistics 1(2):386–411, 2007. https://doi.org/10.1214/07-AOAS122.

[17] J. P. Romano and M. Wolf. Exact and Approximate Stepdown Methods for Multiple Hypothesis Testing. Journal of the American Statistical Association 100(469):94–108, 2005. https://doi.org/10.1198/016214504000000539.

[18] I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White and S.-I. Lee. Learning to Maximize Mutual Information for Dynamic Feature Selection. ICML, PMLR 202:6424–6447, 2023. https://proceedings.mlr.press/v202/covert23a.html.

[19] S. Gadgil, I. Covert and S.-I. Lee. Estimating Conditional Mutual Information for Dynamic Feature Selection. ICLR, 2024. https://arxiv.org/abs/2306.03301.

[20] A. L. I. Norcliffe, C. Lee, F. Imrie, M. van der Schaar and P. Lio. Stochastic Encodings for Active Feature Acquisition. ICML, PMLR 267:46784–46814, 2025. https://proceedings.mlr.press/v267/norcliffe25a.html.

[21] O. B. Guney, K. S. Saichandran, K. Elzokm, Z. Zhang and V. B. Kolachalama. Active Feature Acquisition via Explainability-driven Ranking. ICML, PMLR 267:20748–20765, 2025. https://proceedings.mlr.press/v267/guney25a.html.